3. Requirements Gathering

3.1 Stakeholder Analysis

- **Traffic Management Authorities & City Planners:** Require actionable insights on high-accident locations, hazardous junctions, and peak times to improve urban road infrastructure and optimize patrol deployments.
- **Emergency & Medical Services:** Need accurate metrics on response times, accident severity levels, and road blockage durations (Start_Time to End_Time) to optimize ambulance and rescue team dispatching.
- **General Drivers & Commuters:** Seek analytical safety data regarding severe weather conditions (e.g., fog, precipitation) and dangerous road features (Traffic_Signal, Junction) to plan safer routes.
- **Data Analysts & Project Team:** Require clean, structured data (via Power Query and Star Schema modelling) to build accurate Power BI executive dashboards and predictive models.

3.2 User Stories & Use Cases

- **As a City Planner:** I want to filter accidents by specific states, cities, and streets so that I can identify high-risk hotspots and prioritize road safety improvements.
- **As an Emergency Services Coordinator:** I want to analyze average road blockage duration under various weather conditions so that I can optimize emergency response time and resource allocation.
- **As a Commuter:** I want to understand how weather elements (like temperature and precipitation) and time of day (Night vs. Day) impact accident severity so that I can make safer travel decisions.

3.3 Functional Requirements

ID	Module	Functional Specification

1	Data Ingestion & Cleaning	The system must ingest, clean, and transform large-scale accident datasets using Power Query (handling missing values and converting date-time stamps).
2	Data Modeling	The system must implement a relational Star Schema data model connecting the central accident fact table with dimension tables, Location, Weather, Time).
3	DAX Calculations	The system must calculate dynamic key measures (Total Accidents, Average Severity, Average Blockage Duration).
4	Interactive Dashboards	The system must provide interactive Power BI visualizations allowing users to slice and dice data by severity level, weather conditions, geographical location, and temporal dimensions.

3.4 Non-Functional Requirements

- **Performance:** The data model and dashboards must load and execute calculations efficiently despite the large scale of the national dataset, utilizing query optimization and selective indexing.

- **Usability:** The Power BI user interface must feature an intuitive, user-friendly layout with clear tooltips, simple filtering controls, and consistent color palettes for non-technical stakeholders.
- **Reliability:** The Power Query data pipeline must maintain data integrity, handle null values accurately, and preserve consistent data types across all fields.